



Dynamic stability of cylindrical nanoshells under combined static and periodic axial loads

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Abstract

The dynamic stability of cylindrical nanoshells subjected to combined static and time-dependent periodic axial forces is studied by employing the two-dimensional nonlocal elasticity theory together with the first-order shear deformation theory of shells. The differential quadrature method as an efficient and accurate numerical technique is applied to discretize the equations of motion under different boundary conditions in the spatial domain and transform them into a system of coupled Mathieu–Hill-type equations in time domain. Subsequently, the Bolotin’s first approximation method is employed to extract the dynamic instability regions of the cylindrical nanoshells. The approach is validated by showing its fast convergence rate and carrying out comparison studies with existing results in the limit cases. Afterward, the effects of the nonlocal parameter, length and thickness-to-mean radius ratios together with different boundary conditions on the principal dynamic instability regions of the cylindrical nanoshells are studied in detail.

Keywords Dynamic stability · Cylindrical nanoshells · Nonlocal elasticity theory · Small-scale effect · Differential quadrature · Bolotin’s first approximation

1 Introduction

The high performance and efficiency of micro-/nanoelectromechanical systems (MEMs/NEMs) promise their new technical applications in modern industries. On the other hand, nanostructural elements such as nanobeams, nanoplates and nanoshells are commonly used as the basic components in micro-/nanodevices. It is clear that accurately determining their mechanical behavior is of great importance for the modeling, design and optimization of the MEMs/NEMs.

There are three approaches for the analysis of nanostructures, i.e., atomistic, hybrid atomistic–continuum mechanics and continuum mechanics. The first two approaches are computationally expensive and are not appropriate for analyzing large-scale systems. Furthermore, the classical continuum theory cannot predict the size effect. To overcome these troubles, usually the nonlocal elasticity theory

of Eringen [1–3], which is an approximate model, together with the one-dimensional beam theories and two-dimensional dimensional plates and shell theories, respectively, has been employed to investigate the global mechanical and thermo-mechanical behavior of the nanobeams, nanoplates and nanoshells in recent years; see for example Refs. [4–24].

Up to now, to the best of authors’ knowledge, the only research works concerning the dynamic stability of nanoshells based on the 2D shell theories are those reported by Ghorbanpour Arani et al. [25, 26] and Li and Wang [27]. However, there exist some studies on this topic which have been done by employing the different nonlocal beam theories [28–33].

Ghorbanpour Arani et al. [25] investigated the dynamic stability of simply supported single- and double-walled carbon nanotubes under dynamic axial loading by employing the two-dimensional nonlocal shell theory and the Rayleigh–Ritz method. They considered the effects of the elastic medium and the van der Waals forces between the two layers in the DWCNTs using the Winkler model and Lennard-Jones theory, respectively. In another work, Ghorbanpour Arani et al. [26] studied the dynamic stability of simply supported double-walled boron nitride nanotubes conveying viscous fluid using the nonlocal

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piezoelectricity together with Euler–Bernoulli beam theory, Timoshenko beam theory and cylindrical shell theory. In addition, the surface stress effects were considered based on Gurtin–Murdoch continuum theory. They obtained the dynamic instability region of the system by applying the incremental harmonic balance method. Li and Wang [27] presented the dynamic characteristics of simply supported carbon nanotubes under combined static and periodically oscillating axial loads based on the two-dimensional nonlocal shell theory and a nonlocal effective model of MWCNTs.

In these works, the classical shell theory was used and the influences of the transverse shear deformations and rotary inertia were not considered.

Fu et al. [28] studied the nonlinear dynamic instability of simply supported double-walled nanotubes (DWCNTs) embedded in an elastic medium. They formulated the dynamic governing equations of the carbon nanotubes based on the Euler–Bernoulli beam theory by taking into account the effects of van der Waals forces as well as the non-coaxial curvature of each nested tube. They used the incremental harmonic balanced method to solve the problem.

Ansari et al. [29] illustrated the dynamic stability of simply supported single-walled carbon nanotubes (SWCNTs) under axial compression in a thermal environment. They applied the nonlocal Bernoulli–Euler and Timoshenko beam theories to formulate the problem. They employed a Winkler-type elastic foundation to represent the interaction of the SWCNTs and the surrounding elastic medium.

Ansari et al. [30] examined the dynamic stability of simply supported multi-walled carbon nanotubes (MWCNTs) embedded in an elastic medium subjected to thermal environment based on the nonlocal elastic Timoshenko beam model. They employed a Pasternak-type elastic foundation to represent the interaction between the MWCNT and the surrounding elastic medium.

Hosseini-Hashemi and Ilkhani [31] presented analytical solution for the dynamic stability of a simply supported spinning nanotube under tangential load using Euler and Timoshenko nonlocal beam theories. They obtained equations of motion by applying Hamilton’s principle on the energy terms of nanotube by considering rotational velocity and compressive tangential load.

Karlić et al. [32] investigated the nonlinear vibration and dynamic stability of a simply supported single-walled carbon nanotube embedded in a viscoelastic medium. They derived the governing equations in accordance with Maxwell’s equations and the nonlocal Euler–Bernoulli beam theory under the von Karman nonlinear strain–displacements assumptions. The perturbation method was employed to obtain an approximate analytical solution of nonlinear natural frequency of the nanotubes. Also, they introduced the incremental harmonic balance method to determine the

instability regions for linear and nonlinear cases of dynamic stability.

Saffari et al. [33] investigated dynamic stability of functionally graded (FG) nanobeam under axial and thermal loading with simply supported ends based on the nonlocal Timoshenko beam theory in conjunction with Gurtin–Murdoch continuum theory to implement the surface stress effects. They employed Pasternak elastic medium to represent the interaction of the FG nanobeam and the surrounding elastic medium.

In all of these interesting works, the dynamic stability of the nanotubes with simply supported ends has been investigated based on the different one-dimensional beam theories. But, as the numerical results of this study will show, the short nanoshells cannot be modeled as a beam. On the other hand, for the long nanoshells, the influences of boundary conditions become important and should be investigated. Therefore, a two-dimensional nonlocal theory that can consider both circumferential nanoshell deformation and their transverse flexural behavior becomes necessity for their accurate analysis. In addition, a general method of solution is essential for different boundary conditions implementation. Hence, in this paper, the dynamic stability of nanoshells is presented based on the two-dimensional nonlocal elasticity of Eringen in conjunction with the first-order shear deformation theory (FSDT) of shells. The differential quadrature method (DQM) as an efficient and accurate numerical method [9, 34–44] is utilized to spatially discretize the equations of motion subjected to different boundary conditions. The discretized governing partial differential equations are converted to a system of Mathieu–Hill-type equations. Bolotin’s first approximation method [45] is adopted to extract the dynamic stability behavior of the system from these equations. The convergence of the approach is demonstrated through different examples, and the formulation is validated by performing some comparison studies with existing available numerical data in the literature. Finally, a detailed parametric study is carried out to investigate the influences of nonlocal parameter together with length and thickness-to-mean radius ratios on the principal dynamic instability regions of the nanoshells under different boundary conditions.

2 Governing equations

The system under consideration is a nanoshell which is subjected to an axial compressive time-dependent load $N_{cr}^d(t)$ (Fig. 1). The mean radius, the total thickness and the length of the nanoshell are denoted by R , h and L , respectively. A cylindrical coordinate system on the mid-surface of the shell and with coordinate variables (x, θ, z) is used to label the material points of the nanoshell in its undeformed reference

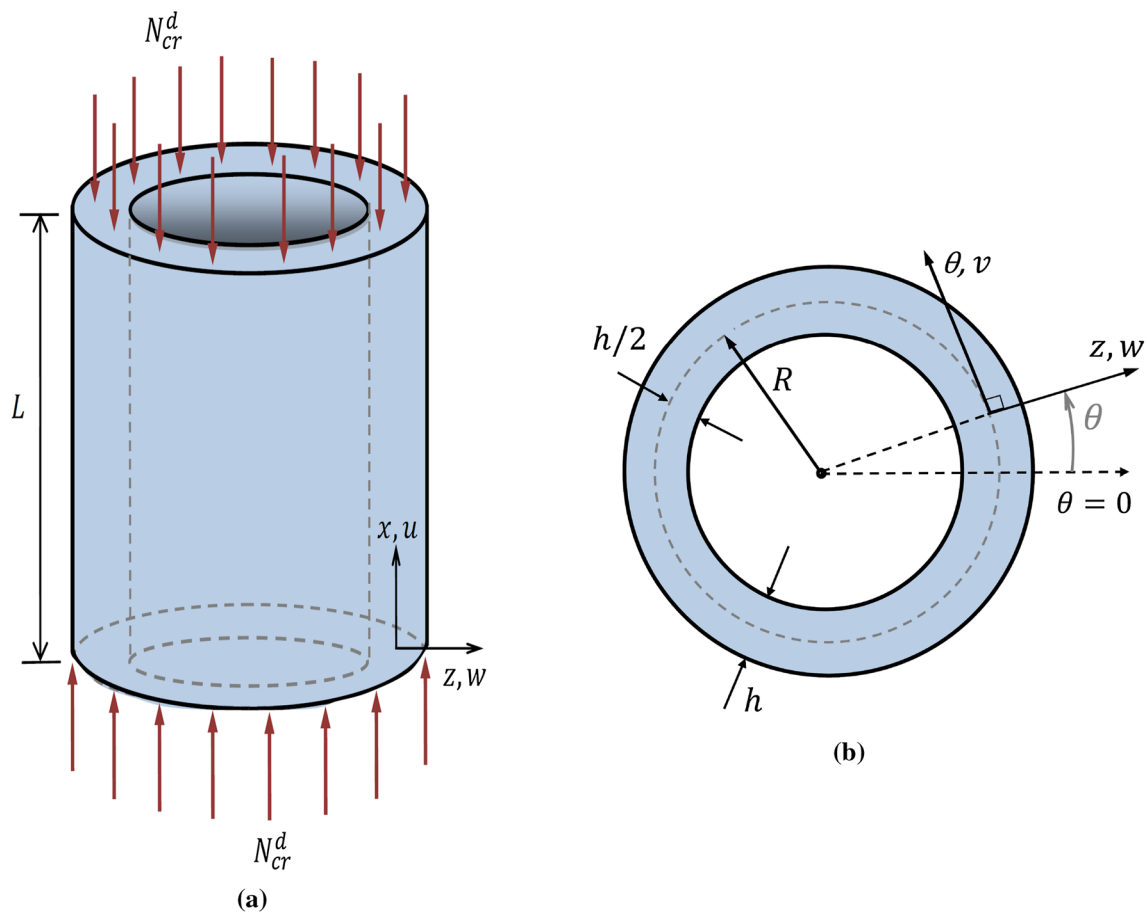


Fig. 1 **a, b** The geometry, coordinate system and loading of the nanoshell

configuration. In the following, the basic governing equations are presented.

Based on the FSDT of shells, the displacement components of a generic material point (x, θ, z) measured from the initial equilibrium state under axial loading along the axial, tangential and radial directions are, respectively, approximated as

$$\begin{aligned}\bar{u}(x, \theta, z, t) &= u(x, \theta, t) + z\varphi^x(x, \theta, t), \quad \bar{v}(x, \theta, z, t) \\ &= v(x, \theta, t) + z\varphi^\theta(x, \theta, t), \quad \bar{w}(x, \theta, z, t) \\ &= w(x, \theta, t),\end{aligned}\quad (1a-c)$$

where u , v and w are the displacement components of the corresponding material point on the mid-surface of the nanoshell along the x -, θ - and z -axes, respectively. Also, φ^x and φ^θ signify the rotation of the normal vector to the mid-surface of the nanoshell at this point about the θ - and x -axes, respectively. As a result, the linear strain components in terms of the displacement and rotation components become

$$\begin{aligned}\epsilon_{xx} &= \frac{\partial u}{\partial x} + z\frac{\partial \varphi^x}{\partial x}, \quad \epsilon_{\theta\theta} = \frac{1}{(R+z)}\left(\frac{\partial v}{\partial \theta} + z\frac{\partial \varphi^\theta}{\partial \theta} + w\right), \\ \gamma_{\theta x} &= \frac{\partial v}{\partial x} + z\frac{\partial \varphi^\theta}{\partial x} + \frac{1}{(R+z)}\left(\frac{\partial u}{\partial \theta} + z\frac{\partial \varphi^x}{\partial \theta}\right), \\ \gamma_{z\theta} &= \varphi^\theta + \frac{1}{(R+z)}\left(\frac{\partial w}{\partial \theta} - v - z\varphi^\theta\right), \quad \gamma_{zx} = \frac{\partial w}{\partial x} + \varphi^x,\end{aligned}\quad (2a-d)$$

where ϵ_{ii} and γ_{ij} are the normal and shear components of linear strain tensor, respectively.

According to the Eringen's nonlocal elasticity theory, the nonlocal constitutive relations for a linear elastic isotropic material are given by [9]

$$(1 - \mu^2 \nabla^2) \sigma_{ij} = \sigma_{ij}^L \quad \text{for } i, j = r, \theta, z, \quad (3)$$

where σ_{ij} and σ_{ij}^L represent the nonlocal and local stress tensor components, respectively; μ is the nonlocal parameter which denotes the small-scale effect; and ∇^2 is the three-dimensional Laplacian operator in cylindrical coordinate system.

Neglecting the transverse normal stress component, the constitutive relations for a thin-to-moderately thick nanoshell can be explained as

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{\theta\theta} \\ \sigma_{x\theta} \\ \sigma_{xz} \\ \sigma_{\theta z} \end{Bmatrix} - \mu^2 \nabla^2 \begin{Bmatrix} \sigma_{xx} \\ \sigma_{\theta\theta} \\ \sigma_{x\theta} \\ \sigma_{xz} \\ \sigma_{\theta z} \end{Bmatrix} = \begin{bmatrix} \frac{E}{(1-\nu^2)} & \frac{\nu E}{(1-\nu^2)} & 0 & 0 & 0 \\ \frac{\nu E}{(1-\nu^2)} & \frac{E}{(1-\nu^2)} & 0 & 0 & 0 \\ 0 & 0 & G & 0 & 0 \\ 0 & 0 & 0 & \kappa_s G & 0 \\ 0 & 0 & 0 & 0 & \kappa_s G \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{\theta\theta} \\ \gamma_{x\theta} \\ \gamma_{xz} \\ \gamma_{\theta z} \end{Bmatrix}, \quad (4)$$

where E , G and ν are Young's modulus, shear rigidity and Poisson's ratio of the nanoshell, respectively; also, κ_s is the shear correction factor of the nanoshell.

In order to derive the equations of motion together with the related boundary conditions in a systematic manner, Hamilton's principle is employed:

$$\int_{t_1}^{t_2} (\delta K - \delta U + \delta W) dt = 0, \quad (5)$$

where t_i ($i = 1, 2$) are two arbitrary times, δ is the variational operator, K and U are the kinetic and potential energies of the nanoshell, respectively; also, W represents the work of external load.

Based on the FSDT of shells, the variation of the potential and kinetic energies of the nanoshell takes the following form, respectively:

$$\delta U = \int_0^{2\pi} \int_0^L \int_{-\frac{h}{2}}^{\frac{h}{2}} [\sigma_{xx} \delta \varepsilon_{xx} + \sigma_{\theta\theta} \delta \varepsilon_{\theta\theta} + \sigma_{xz} \delta \gamma_{xz} + \sigma_{x\theta} \delta \gamma_{x\theta} + \sigma_{\theta z} \delta \gamma_{\theta z}] (R+z) dz dx d\theta \quad (6)$$

$$\delta K = \int_0^{2\pi} \int_0^L \int_{-\frac{h}{2}}^{\frac{h}{2}} \rho \left[\frac{\partial u}{\partial t} \frac{\partial \delta u}{\partial t} + \frac{\partial v}{\partial t} \frac{\partial \delta v}{\partial t} + \frac{\partial w}{\partial t} \frac{\partial \delta w}{\partial t} + z^2 \frac{\partial \varphi^x}{\partial t} \frac{\partial \delta \varphi^x}{\partial t} + z^2 \frac{\partial \varphi^\theta}{\partial t} \frac{\partial \delta \varphi^\theta}{\partial t} \right] (R+z) dz dx d\theta, \quad (7)$$

where ρ is the mass density of the nanoshell.

Also, the virtual work of the in-plane load N_{cr}^d can be expressed as

$$\delta W = \int_0^{2\pi} \int_0^L \int_{-\frac{h}{2}}^{\frac{h}{2}} N_{cr}^d \frac{\partial w}{\partial x} \frac{\partial \delta w}{\partial x} (R+z) dz dx d\theta. \quad (8)$$

By substituting Eqs. (2), (4) and (6)–(8) into Eq. (5) and performing the integration by parts and some other manipulations, the governing differential equations and the related boundary conditions can be extracted as

Governing differential equations

$$\begin{aligned} \delta u : & A_1^n \frac{\partial^2 u}{\partial x^2} + \bar{A}_{-1}^m \frac{\partial^2 u}{\partial \theta^2} + (\bar{A}_0^m + \bar{A}_0^n) \frac{\partial^2 v}{\partial \theta \partial x} \\ & + \bar{A}_0^n \frac{\partial w}{\partial x} + B_1^n \frac{\partial^2 \varphi^x}{\partial x^2} + \bar{B}_{-1}^m \frac{\partial^2 \varphi^x}{\partial \theta^2} + (\bar{B}_0^n + \bar{B}_0^m) \frac{\partial^2 \varphi^\theta}{\partial \theta \partial x} \\ & = I_{11} \frac{\partial^2 u}{\partial t^2} + I_{12} \frac{\partial^2 \varphi^x}{\partial t^2} - \mu^2 \nabla^2 \left(I_{11} \frac{\partial^2 u}{\partial t^2} + I_{12} \frac{\partial^2 \varphi^x}{\partial t^2} \right) \end{aligned} \quad (9)$$

$$\begin{aligned} \delta v : & (\bar{A}_0^n + \bar{A}_0^m) \frac{\partial^2 u}{\partial \theta \partial x} + \bar{A}_1^m \frac{\partial^2 v}{\partial x^2} \\ & + A_{-1}^n \frac{\partial^2 v}{\partial \theta^2} - A_{-1}^m v + (A_{-1}^m + A_{-1}^n) \frac{\partial w}{\partial \theta} \\ & + (\bar{B}_0^n + \bar{B}_0^m) \frac{\partial^2 \varphi^x}{\partial \theta \partial x} + \bar{B}_1^m \frac{\partial^2 \varphi^\theta}{\partial x^2} \\ & + B_{-1}^n \frac{\partial^2 \varphi^\theta}{\partial \theta^2} + (A_0^m - B_{-1}^m) \varphi^\theta \\ & = I_{11} \frac{\partial^2 v}{\partial t^2} + I_{12} \frac{\partial^2 \varphi^\theta}{\partial t^2} \\ & - \mu^2 \nabla^2 \left(I_{11} \frac{\partial^2 v}{\partial t^2} + I_{12} \frac{\partial^2 \varphi^\theta}{\partial t^2} \right) \end{aligned} \quad (10)$$

$$\begin{aligned} \delta w : & -\bar{A}_0^n \frac{\partial u}{\partial x} - (A_{-1}^n + A_{-1}^m) \frac{\partial v}{\partial \theta} + A_1^m \frac{\partial^2 w}{\partial x^2} \\ & + A_{-1}^m \frac{\partial^2 w}{\partial \theta^2} - A_{-1}^n w + (A_1^m - \bar{B}_0^n) \frac{\partial \varphi^x}{\partial x} \\ & + (A_0^m - B_{-1}^m - B_{-1}^n) \frac{\partial \varphi^\theta}{\partial \theta} - \hat{N}_{xx} \frac{\partial^2 w}{\partial x^2} \\ & = I_{11} \frac{\partial^2 w}{\partial t^2} - \mu^2 \nabla^2 \left(I_{11} \frac{\partial^2 w}{\partial t^2} \right) \end{aligned} \quad (11)$$

$$\begin{aligned} \delta \varphi^x : & B_1^n \frac{\partial^2 u}{\partial x^2} + \bar{B}_{-1}^m \frac{\partial^2 u}{\partial \theta^2} + (\bar{B}_0^n + \bar{B}_0^m) \frac{\partial^2 v}{\partial x \partial \theta} \\ & + (B_0^n - A_1^m) \frac{\partial w}{\partial x} + D_1^n \frac{\partial^2 \varphi^x}{\partial x^2} + \bar{D}_{-1}^m \frac{\partial^2 \varphi^x}{\partial \theta^2} - A_1^m \varphi^x \\ & + (\bar{D}_0^n + \bar{D}_0^m) \frac{\partial^2 \varphi^\theta}{\partial x \partial \theta} = I_{12} \frac{\partial^2 v}{\partial t^2} + I_{13} \frac{\partial^2 \varphi^\theta}{\partial t^2} \\ & - \mu^2 \nabla^2 \left(I_{12} \frac{\partial^2 v}{\partial t^2} + I_{13} \frac{\partial^2 \varphi^\theta}{\partial t^2} \right) \end{aligned} \quad (12)$$

$$\begin{aligned}
\delta\varphi^\theta : & (\bar{B}_0^n + \bar{B}_0^m) \frac{\partial^2 u}{\partial x \partial \theta} + \bar{B}_1^m \frac{\partial^2 v}{\partial x^2} + B_{-1}^n \frac{\partial^2 v}{\partial \theta^2} \\
& + (A_0^m - B_{-1}^m) v + (B_{-1}^n + B_{-1}^m - A_0^m) \frac{\partial w}{\partial \theta} \\
& + (\bar{D}_0^n + \bar{D}_0^m) \frac{\partial^2 \varphi^x}{\partial x \partial \theta} + \bar{D}_1^m \frac{\partial^2 \varphi^\theta}{\partial x^2} \\
& + D_{-1}^n \frac{\partial^2 \varphi^\theta}{\partial \theta^2} + (2B_0^m - D_{-1}^m - A_1^m) \varphi^\theta \quad (13) \\
= & I_{12} \frac{\partial^2 u}{\partial t^2} + I_{13} \frac{\partial^2 \varphi^x}{\partial t^2} \\
& - \mu^2 \nabla^2 \left(I_{12} \frac{\partial^2 u}{\partial t^2} + I_{13} \frac{\partial^2 \varphi^x}{\partial t^2} \right)
\end{aligned}$$

Boundary conditions at the ends $x = 0$ and $x = L$:

Either $\delta u = 0$ or

$$A_1^n \frac{\partial u}{\partial x} + \bar{A}_0^n \frac{\partial v}{\partial \theta} + \bar{A}_0^m w + B_1^n \frac{\partial \varphi^x}{\partial x} + \bar{B}_0^n \frac{\partial \varphi^\theta}{\partial \theta} = 0 \quad (14a,b)$$

Either $\delta v = 0$ or

$$\bar{A}_0^n \frac{\partial u}{\partial \theta} + \bar{A}_1^m \frac{\partial v}{\partial x} + \bar{B}_0^m \frac{\partial \varphi^x}{\partial \theta} + \bar{B}_1^m \frac{\partial \varphi^\theta}{\partial x} = 0 \quad (15a,b)$$

Either $\delta w = 0$ or

$$A_1^m \left(\varphi^x + \frac{\partial w}{\partial x} \right) = 0 \quad (16a,b)$$

Either $\delta \varphi^x = 0$ or

$$B_1^n \frac{\partial u}{\partial x} + \bar{B}_0^n \frac{\partial v}{\partial \theta} + \bar{B}_0^m w + D_1^n \frac{\partial \varphi^x}{\partial x} + \bar{D}_0^n \frac{\partial \varphi^\theta}{\partial \theta} = 0 \quad (17a,b)$$

Either $\delta \varphi^\theta = 0$ or

$$\bar{B}_0^n \frac{\partial u}{\partial \theta} + \bar{B}_1^m \frac{\partial v}{\partial x} + \bar{D}_0^m \frac{\partial \varphi^x}{\partial \theta} + \bar{D}_1^m \frac{\partial \varphi^\theta}{\partial x} = 0, \quad (18a,b)$$

where $\hat{N}_{xx} = \int_{-\frac{h}{2}}^{\frac{h}{2}} N_{cr}^d (R+z) dz$; also, the stiffness coefficients and the inertia terms appeared in the above equations are, respectively,

$$\begin{aligned}
\begin{bmatrix} A_i^n & B_i^n & D_i^n & L_i^n \\ \bar{A}_i^n & \bar{B}_i^n & \bar{D}_i^n & \bar{L}_i^n \end{bmatrix} &= \int_{-\frac{h}{2}}^{\frac{h}{2}} \frac{E}{(1-\nu^2)} \begin{bmatrix} 1 \\ z \\ z^2 \\ z^3 \end{bmatrix} \begin{bmatrix} 1 \\ z \\ z^2 \\ z^3 \end{bmatrix} (R+z)^i dz, \\
\begin{bmatrix} A_i^m & B_i^m & D_i^m \\ \bar{A}_i^m & \bar{B}_i^m & \bar{D}_i^m \end{bmatrix} &= \int_{-\frac{h}{2}}^{\frac{h}{2}} G \begin{bmatrix} \kappa_s \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ z \\ z^2 \end{bmatrix} (R+z)^i dz \quad \text{for } i = -1, 0, 1 \quad (19a,b)
\end{aligned}$$

$$(I_{11}, I_{12}, I_{13}) = \int_{-\frac{h}{2}}^{\frac{h}{2}} \rho (1, z, z^2) (R+z) dz. \quad (20)$$

In the present work, cylindrical nanoshells with hard simply support (S), clamped (C) and combinations of these boundary conditions are analyzed. The geometrical and natural conditions of these constraints are as follows:

Hard simply support (S):

$$\begin{aligned}
u = 0, \quad v = 0, \quad w = 0, \quad B_1^n \frac{\partial u}{\partial x} + \bar{B}_0^n \frac{\partial v}{\partial \theta} + \bar{B}_0^m w + D_1^n \frac{\partial \varphi^x}{\partial x} + \bar{D}_0^n \frac{\partial \varphi^\theta}{\partial \theta} = 0, \\
\bar{B}_0^n \frac{\partial u}{\partial \theta} + \bar{B}_1^m \frac{\partial v}{\partial x} + \bar{D}_0^m \frac{\partial \varphi^x}{\partial \theta} + \bar{D}_1^m \frac{\partial \varphi^\theta}{\partial x} = 0. \quad (21a-e)
\end{aligned}$$

Clamped (C):

$$u = 0, \quad v = 0, \quad w = 0, \quad \varphi^x = 0, \quad \varphi^\theta = 0. \quad (22a-e)$$

Due to the geometrical periodicity of the cylindrical nanoshells, the circumferential dependence of the equations of motion and the boundary conditions can be eliminated if one considers the displacement and rotation components in the following form:

$$\begin{aligned}
\begin{Bmatrix} u(x, \theta, t) \\ w(x, \theta, t) \\ \varphi^x(x, \theta, t) \end{Bmatrix} &= \begin{Bmatrix} U_m(x, t) \\ W_m(x, t) \\ \Phi_m^x(x, t) \end{Bmatrix} \cos(m\theta), \\
\begin{Bmatrix} v(x, \theta, t) \\ \varphi^\theta(x, \theta, t) \end{Bmatrix} &= \begin{Bmatrix} V_m(x, t) \\ \Phi_m^\theta(x, t) \end{Bmatrix} \sin(m\theta), \quad (23a,b)
\end{aligned}$$

where $m (= 0, 1, 2, \dots, \infty)$ is the circumferential wave number of the nanoshell. After substituting these functions into the equations of motion and the related boundary conditions, the result is a system of time-dependent one-dimensional differential equations and, in the following section, their solution procedure is described.

3 Solution procedure

Since it is difficult to solve analytically the coupled equations of motion under different boundary conditions, the differential quadrature method (DQM) as an accurate and low computational efforts numerical tool [9, 34–44] is applied to discretize the equations of motion and the related boundary conditions in the spatial domain. The main advantage of this method is that the equations of motion and the boundary conditions are discretized in their strong form and only limited number of grid points is required to achieve accurate converged results. Based on this method, the first- and second-order derivatives of a typical function $f(x, t)$ with respect to coordinate variable x at the grid point x_i are approximated as

$$\left. \frac{\partial f(x, t)}{\partial x} \right|_{x=x_i} = \sum_{j=1}^{N_x} A_{ij}^x f(x_j, t) = \sum_{j=1}^{N_x} A_{ij}^x f_j, \quad \left. \frac{\partial^2 f(x, t)}{\partial x^2} \right|_{x=x_i} = \sum_{j=1}^{N_x} B_{ij}^x f_j, \quad (24a,b)$$

where A_{ij}^x and B_{ij}^x are the weighting coefficients associated with the first- and second-order along the x -axis. One can realize from Eq. (24) that the weighting coefficients and grid

distribution are two important aspects that control the accuracy of the DQM [36]. The uniformly distributed grid points are a simple and natural grid generation rule. But, it has been shown that the roots of orthogonal polynomials such as the Chebyshev polynomials are a better choice. Hence, in this study, the roots of the Chebyshev polynomials are adopted to generate the grid points [36], which are

$$x_i = \frac{L}{2} \left\{ 1 - \cos \left[\frac{(i-1)\pi}{(N_x-1)} \right] \right\} \quad \text{for } i = 1, 2, \dots, N_x. \quad (25)$$

In order to calculate the weighting coefficients, a set of test functions should be inserted in Eq. (24). By using the Lagrange polynomials, Shu and Richards [46] developed explicit formulation for the weighting coefficients [36] and these formulas are used in this work:

$$A_{ij}^x = \begin{cases} \frac{M(x_i)}{(x_i - x_j)} & i \neq j \\ -\sum_{i \neq j}^N A_{ij}^x & i = j \end{cases}, \quad i, j = 1, 2, \dots, N_x; \quad [B_{ij}^x] = [A_{ij}^x]^2, \quad (26)$$

where $M(x_i) = \prod_{j=1, j \neq i}^{N_x} (x_i - x_j)$.

By employing the DQM discretization rules, the equations of motion and the corresponding boundary conditions are transformed to a set of ordinary differential equations in temporal domain. For brevity, the details of the discretization procedure are not presented here and only the discretized form of Eq. (9) is given as an example. Based on the DQM, at a domain grid point x_i with $i = 2, \dots, N_x - 1$, this equation takes the following form:

$$\begin{aligned} & A_1^n \sum_{k=1}^{N_x} B_{ik}^x U_{mk} - m^2 \bar{A}_{-1}^m U_{mi} + m(\bar{A}_0^m + \bar{A}_0^n) \\ & \sum_{k=1}^{N_x} A_{ik}^x V_{mk} + \bar{A}_0^n \sum_{k=1}^{N_x} A_{ik}^x W_{mk} + B_1^n \sum_{k=1}^{N_x} B_{ik}^x \Phi_{mk}^x - m^2 \bar{B}_{-1}^m \Phi_{mi}^x \\ & + (\bar{B}_0^m + \bar{B}_0^n) \sum_{k=1}^{N_x} A_{ik}^x \Phi_{mk}^0 = I_{11} \frac{d^2 U_{mi}}{dt^2} + I_{12} \frac{d^2 \Phi_{mi}^x}{dt^2} \quad (27) \\ & - \mu^2 \left(-m^2 \hat{I}_{11} \frac{d^2 U_{mi}}{dt^2} + I_{11} \sum_{k=1}^{N_x} B_{ik}^x \frac{d^2 U_{mk}}{dt^2} \right. \\ & \left. - m^2 \hat{I}_{12} \frac{d^2 \Phi_{mi}^x}{dt^2} + I_{12} \sum_{k=1}^{N_x} B_{ik}^x \frac{d^2 \Phi_{mk}^x}{dt^2} \right), \end{aligned}$$

where $(\hat{I}_{11}, \hat{I}_{12}) = \int_{-\frac{h}{2}}^{\frac{h}{2}} \rho(1, z)(R+z)^{-1} dz$.

To reduce the computational efforts, one can easily eliminate the boundary degrees of freedom from the discretized equations of motion by using the discretized boundary conditions [35]. After doing this, the resulting system of linear

Table 1 Convergence of the non-dimensional excitation frequency (λ) exhibiting the lower bond of boundaries of the principle instability region (i.e., $\beta=0$) of the nanoshells [$h/R=0.05$, $L/R=2.5$, $\alpha=0.25$]

m	N_x	μ (nm)			
		0	0.05	0.1	0.2
C-C					
0	15	1.5542	1.5511	1.5421	1.5073
	17	1.5540	1.5510	1.5419	1.5072
	19	1.5540	1.5509	1.5419	1.5071
	21	1.5540	1.5509	1.5419	1.5071
1	15	1.3059	1.3016	1.2889	1.2415
	17	1.3057	1.3014	1.2887	1.2414
	19	1.3057	1.3014	1.2887	1.2413
	21	1.3057	1.3014	1.2887	1.2413
2	15	1.2021	1.1937	1.1693	1.0847
	17	1.2020	1.1935	1.1691	1.0845
	19	1.2020	1.1935	1.1691	1.0845
	21	1.2020	1.1935	1.1691	1.0845
S-S					
0	13	1.5539	1.5509	1.5418	1.5071
	15	1.5535	1.5504	1.5413	1.5066
	17	1.5535	1.5504	1.5413	1.5066
1	13	1.2985	1.2944	1.2821	1.2361
	15	1.2981	1.2939	1.2816	1.2357
	17	1.2981	1.2939	1.2816	1.2357
2	13	1.1936	1.1853	1.1615	1.0788
	15	1.1934	1.1852	1.1613	1.0787
	17	1.1934	1.1852	1.1613	1.0787

ordinary differential equations in the matrix form can be written as

$$[M] \{\ddot{D}\} + ([K] - N_{cr}^d(t)[G])\{D\} = \{0\}, \quad (28)$$

where $\{D\}$ is the vector of domain degrees of freedom and a superposed dot means differentiation with respect to time; also, the elements of the mass matrix $[M]$, the stiffness matrix $[K]$ and the geometric stiffness matrix $[G]$ are obtained from the discretized form of the equations of motion after imposing the related boundary conditions.

It is assumed that the time-varying axial load is periodic and is expressed in the following form:

$$N_{cr}^d(t) = (\alpha + \beta \cos \omega t) N_{cr}, \quad (29)$$

Table 2 Accuracy of the natural frequencies (Hz) for a soft simply supported homogeneous cylindrical shell ($h/R=0.05$, $L/R=20$)

m	Present	Loy et al. [47]
1	12.898	12.917
2	31.543	31.603
3	87.978	88.267
4	167.946	168.99
5	270.392	273.14

Table 3 Accuracy of the natural frequencies (THz) for a soft simply supported nanotube [$R=5$ (nm), $\nu=0.25$, $L/R=1$, $m=1$, $\mu=0.5$ (nm), $Eh=360$ (J/m²)]

ρ (kg/m ³)	Present	Rahmanian et al. [48]
2150	1.0892	1.1001
2300	1.0531	1.0635
2700	0.9720	0.9816

Table 4 Convergence and accuracy of the non-dimensional buckling load ($\lambda_{cr} = \tilde{N}_{cr} \frac{R^2}{Eh^3}$) of a simply supported homogeneous cylindrical shell subjected to static compressive axial load ($h/R=0.1$, $L/R=5$, $\nu=0.3$)

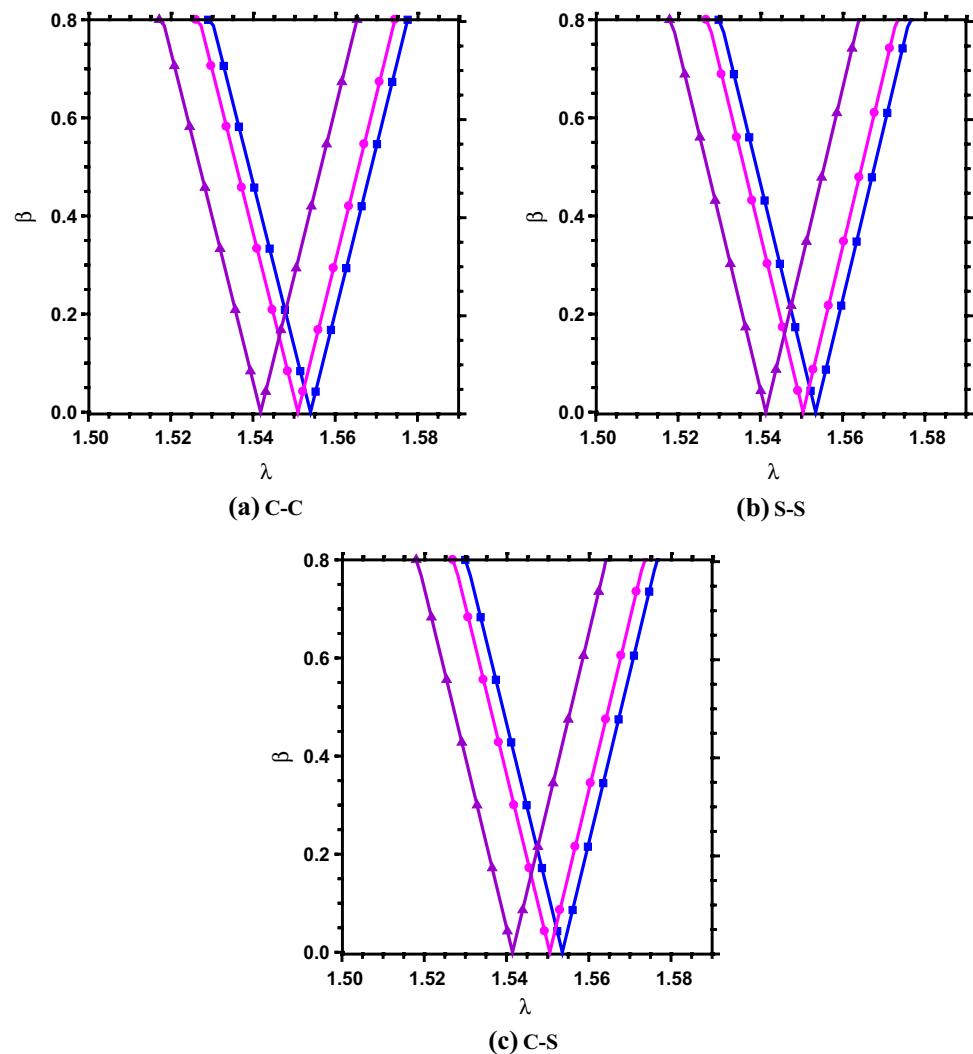
$N_x=17$	19	21	25	27	Timoshenko and Gere [49]
6.124	6.039	6.035	6.022	6.022	6.053

where α and β are, respectively, the static and dynamic load factors, N_{cr} is the critical static axial buckling load of the nanoshell, and also, α denotes the excitation frequency. The critical static axial buckling load N_{cr} is obtained from Eq. (28) by setting $N_{cr}^d(t) = N_{cr}$ and neglecting the inertia terms

$$([K] - N_{cr}[G])\{D\} = \{0\}. \quad (30)$$

On substituting Eq. (29) into Eq. (28), the result is a second-order differential equation with periodic coefficients. It

is clear that this equation is the Mathieu–Hill-type equation and, hence, its regions of unstable solutions are separated by periodic solutions having period T and $2T$ with $T = 2\pi/\omega$. Usually, the solution with period $2T$ has larger unstable regions than the solution with period T . Hence, this solution becomes practically more important than the solution with period T . Using the Bolotin's approach [45] and as a first-order approximation, the solutions with periods $2T$ are determined from the following equation

Fig. 2 a–c Small-scale effect on the instability region of the nanoshells [$h/R=0.05$, $L/R=2.5$, $m=0$, $\alpha=0.25$, $N_x=21$]: (solid line with square) $\mu=0$ (nm), (solid line with circle) $\mu=0.05$ (nm), (solid line with triangle) $\mu=0.1$ (nm)

$$\left| \begin{pmatrix} [K] - \left(\alpha - \frac{\beta}{2}\right)N_{cr}[G] & 0 \\ 0 & [K] - \left(\alpha + \frac{\beta}{2}\right)N_{cr}[G] \end{pmatrix} - \omega^2 \begin{pmatrix} \frac{1}{4}[M] & 0 \\ 0 & \frac{1}{4}[M] \end{pmatrix} \right| = 0. \quad (31)$$

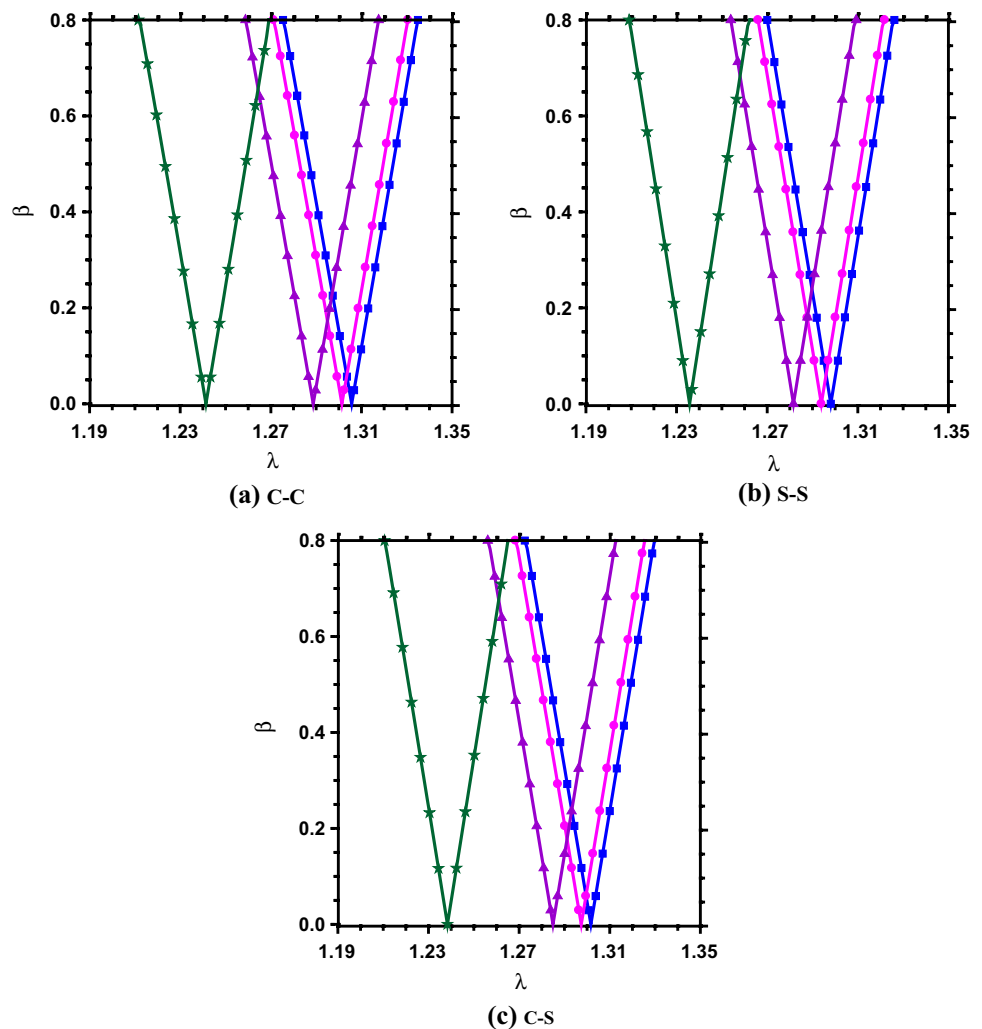
In order to plot the variation of the excitation frequency ω versus the dynamic load factor (β) for a given value of α , firstly, the critical buckling load N_{cr} of the same nanoshell must be extracted from Eq. (30). After that, by solving the system of eigenvalue Eq. (31) for the obtained value of α and a given value of β , the excitation frequency ω is determined. The variation of the lowest frequency ω obtained from Eq. (31) versus β presents the principal instability regions of the nanoshells under the action of the periodic axial compressive load.

4 Numerical results

In this section, at first by showing the fast rate of convergence and performing comparison studies with the other existing results in the open literature, the computational efficiency and accuracy of the method of solution and also the correctness of the formulation are established. Then, the effects of the small-scale effect together with the geometrical parameters on the principal instability regions of the cylindrical nanoshells under different boundary conditions are investigated. For convenience of representing the numerical results, the non-dimensional excitation frequency parameter λ is defined to be

$$\lambda = \omega R \sqrt{\frac{\rho(1-\nu^2)}{E}}. \quad (32)$$

Fig. 3 a–c Small-scale effect on the instability region of the nanoshells [$h/R=0.05$, $L/R=2.5$, $m=1$, $\alpha=0.25$, $N_x=21$]: (solid line with square) $\mu=0$ (nm), (solid line with circle) $\mu=0.05$ (nm), (solid line with triangle) $\mu=0.1$ (nm), (solid line with asterisk) $\mu=0.2$ (nm)



In the numerical computations, otherwise specified, the Poisson's ratio and the shear correction factor are taken to be, respectively, $\nu = 0.25$ and $\kappa_s = 5/6$.

As a first example, the fast convergence behavior and the numerical stability of the method for determining the non-dimensional excitation frequency (λ) of the nanoshells, which shows the lower boundaries of the principal instability region (i.e., $\beta = 0$), is demonstrated in Table 1. In this table, the results for the nanoshells with both ends clamped (C–C) and hard simply supported (S–S) and for different values of the circumferential wave number (m) and small-scale parameter (μ) are presented.

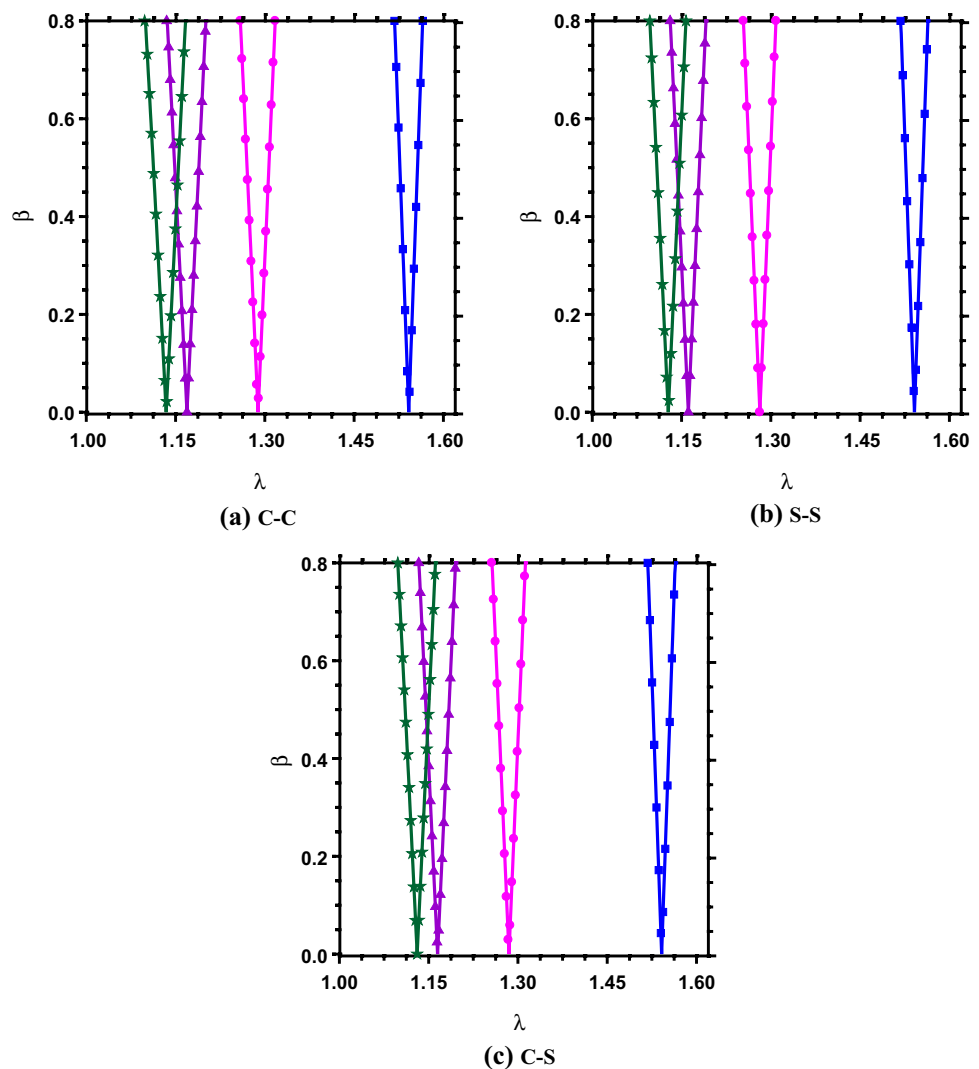
In order to validate the accuracy of the presented method, as a limit case, the natural frequencies of a soft simply supported homogeneous cylindrical shell are compared with those obtained by Loy et al. [47] in Table 2. The results in Ref. [47] have been extracted based on the Love's shell theory by using the Rayleigh–Ritz method. It can be observed that excellent agreement exists between the results of the two

approaches for different values of the circumferential wave number (m). Also, it can be seen that the obtained fundamental natural frequencies are slightly smaller than those of Ref. [47], which can validate more the present approach. These small differences are due to two facts. Firstly, the FSDT includes the transverse shear deformations, which consequently reduce the overall stiffness of the shell with respect to the classical shell theory used by Loy et al. [47]. Secondly, the Rayleigh–Ritz method gives the upper bond of the fundamental natural frequencies.

As another check case, the fundamental frequency of a nanoshell with soft simply supported ends is compared with those obtained by Rahmanian et al. [48] in Table 3. The results, which are obtained for different values of mass density (ρ) of the nanoshell, are in close agreement with those of Ref. [48].

As another comparison study to verify the accuracy of the present approach for the buckling analysis of cylindrical shells, a hard simply supported homogeneous cylindrical

Fig. 4 a–c The effect of the different circumferential wave numbers on the instability region of the nanoshells [$h/R=0.05$, $L/R=2.5$, $\alpha=0.25$, $\mu=0.1$ (nm)]: (solid line with square) $m=0$, (solid line with circle) $m=1$, (solid line with triangle) $m=2$, (solid line with asterisk) $m=3$



shell subjected to uniform static compressive axial loading is considered. The determined critical buckling load is compared with the analytical solution of Timoshenko and Gere [49] in Table 4. They obtained an analytical solution for the critical buckling load of an intermediate length homogeneous cylindrical shell with hard simply supported ends as

$$\bar{N}_{cr} = \frac{Eh^2}{R\sqrt{3(1-\nu^2)}}. \quad (33)$$

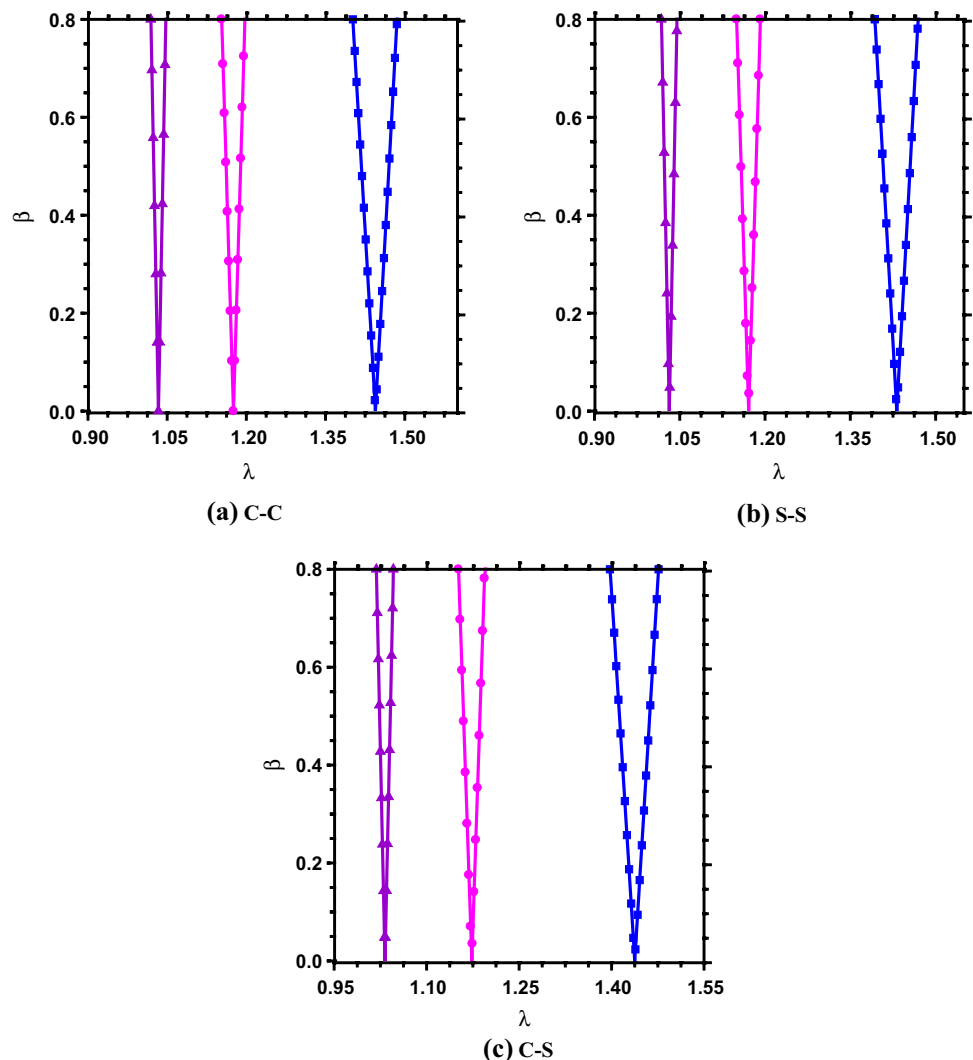
Close agreement between the two sets of results indicates the accuracy of the present approach.

The effect of the small-scale parameter on the principal dynamic instability regions of the nanoshells with different sets of boundary conditions is shown in Figs. 2 and 3 for the circumferential wave number $m=0$ and $m=1$, respectively. It can be seen that in all cases by increasing the small-scale parameter, the onset of instable region decreases. This

means that for a given value of the frequency, the dynamic instability occurs at a lower axial dynamic load. In addition, due to the effect of reducing the small-scale parameter, the region of the dynamic instability is almost the same, but it delays the beginning of the dynamic instability by shifting this region to the higher frequency values.

The principal dynamic instability regions of the nanoshells corresponding to different circumferential wave numbers (m) are shown in Fig. 4. The results are prepared for nanoshells subjected to three different sets of boundary conditions. It is interesting to note that in spite of the nanoshell boundary conditions, by increasing the circumferential wave number m , the corresponding onset of instability region decreases. This indicates that when the excitation frequency is increased from zero, the dynamic instability occurs at asymmetric mode (i.e., a mode with $m \neq 0$) firstly. On the other hand, if the dynamic load factor (β) is increased from zero at a given high excitation frequency, the dynamic instability may happen at symmetric mode $m=0$. Also, by

Fig. 5 a–c The effect of the length-to-mean radius ratio on the instability region of the nanoshells [$h/R=0.05$, $m=1$, $\alpha=0.25$, $\mu=0.1$ (nm)]: (solid line with square) $L/R=2$, (solid line with circle) $L/R=3$, (solid line with triangle) $L/R=4$

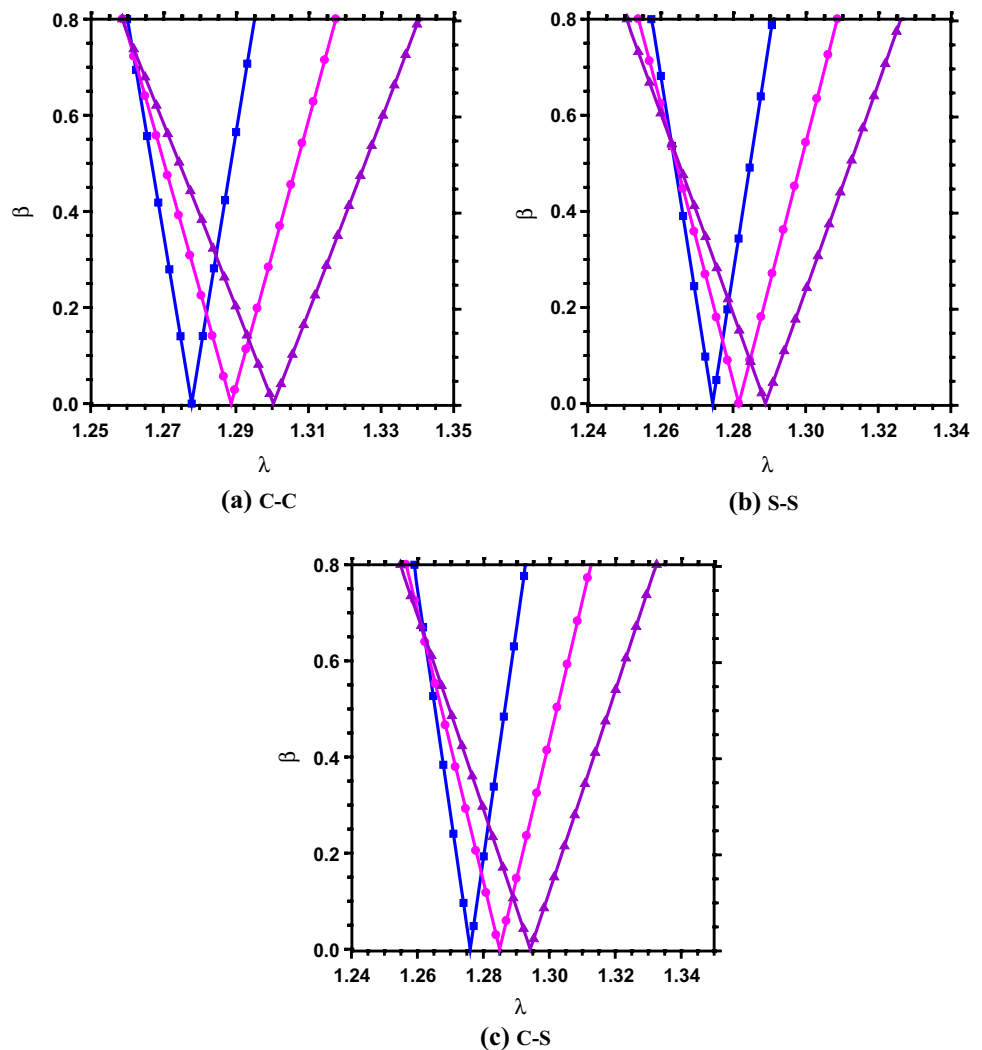


increasing the circumferential wave number m , the differences between instability regions of the nanoshells decrease for the four types of boundary conditions. One can see that the instability regions are slightly affected by the circumferential wave number m .

The use of shell models for the dynamic stability of nanoshells is usually more complicated than the use of beam models, but shell models allow the analysis of nanoshells with small length-to-mean radius ratio. For these structures, the use of 1D beam models gives significant errors because short nanoshells are not one-dimensional structures. Two-dimensional shell models, such as FSDT of shells, are suitable for the correct dynamic instability analysis of short nanoshells, in particular when “thickness-to-mean radius” ratio is small. Therefore, to illustrate this issue, the influences of the length- and thickness-to-mean radius ratios, as important geometric parameters, on the principal dynamic instability region of the nanoshells are shown in Figs. 5 and 6, respectively. It can be seen that these geometrical parameters have considerable effects on the principal dynamic

instability regions of the nanoshells. This study is also performed for the nanoshells with three different boundary conditions. In these figures, the values of the length-to-mean radius ratio (L/R) are in the range of 2–4 (short nanoshell) and the thickness-to-mean radius ratio (h/R) varies between 0.03 and 0.07 (thin nanoshell) which requests the use of 2D shell models. Some difficulties may arise when beam models are used for the analysis of such nanoshells. The use of 2D shell models, such as FSDT theory, overcomes this problem. Specially, it is obvious that by increasing the length-to-mean radius ratio, the bifurcation excitation frequency and the region of the dynamic instability decrease. But, when increasing the thickness-to-mean radius ratio, the bifurcation excitation frequency and the region of the dynamic instability increase. Also, it can be seen that regardless of end boundary conditions of the nanoshell, the trend of variation of the dynamic instability regions versus these geometrical parameters is almost the same. In overall, it can be concluded that both of these geometrical parameters have

Fig. 6 a–c The effect of the thickness-to-mean radius ratio on the instability region of the nanoshells [$L/R=2.5$, $m=1$, $\alpha=0.25$, $\mu=0.1$ (nm)]: (solid line with square) $h/R=0.03$, (solid line with circle) $h/R=0.05$, (solid line with triangle) $h/R=0.07$



considerable effect on the dynamic stability behavior of the nanoshell.

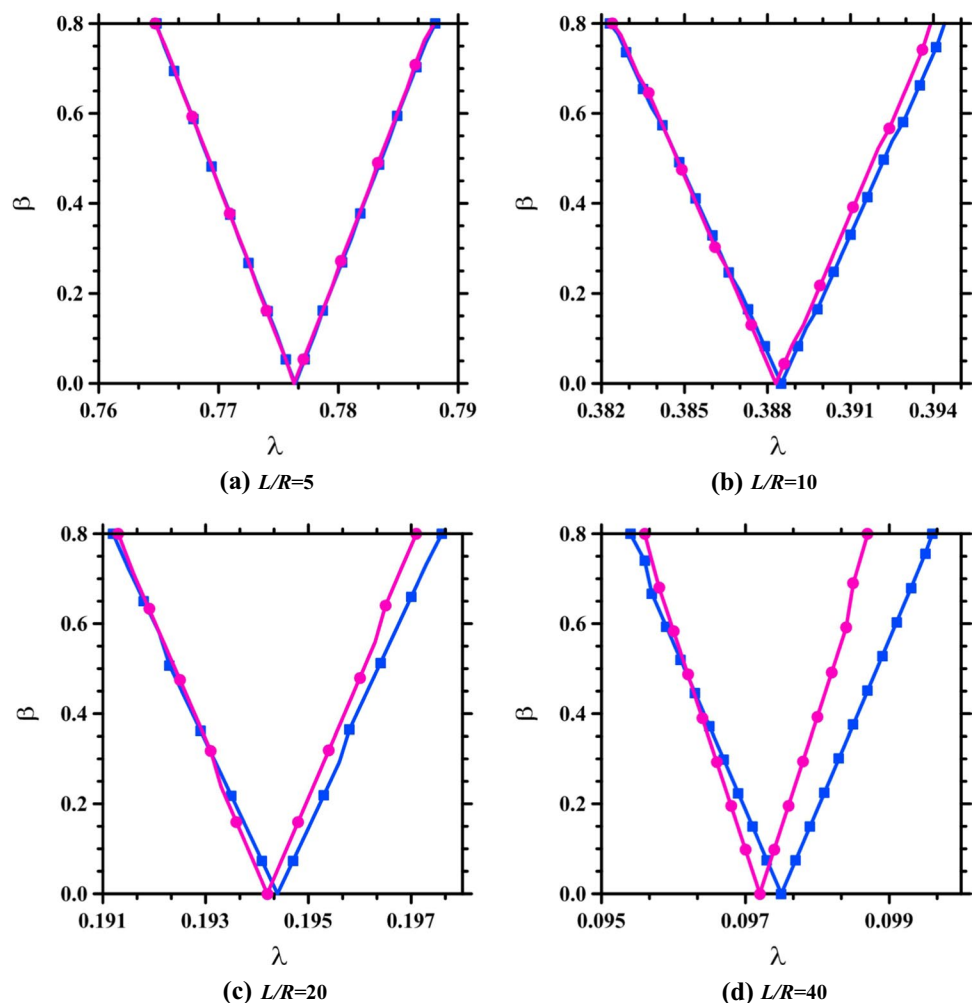
In Fig. 7a–d, the effects of the different length-to-mean radius ratios on the principal dynamic instability region of the nanoshell with clamped and simply supported ends are compared. It can be seen that by increasing the length-to-mean radius ratio of the nanoshell, the differences between the principal dynamic instability regions of the nanoshell for two boundary conditions increase. This is because for the short nanoshells (i.e., nanoshells with approximately $L/R \leq 2.5$), the circumferential deformations of the nanoshell become predominant and it acts as a shell. But for the long nanoshells (i.e., nanoshells with approximately $L/R \geq 2.5$), the transverse flexural behavior of the nanoshell becomes predominant and the nanoshell acts as a beam. Hence, for the short nanoshells, the differences between the influences of the clamped and simply supported are low, but, for the long nanoshells, the effects of the boundary

conditions become important and boundary conditions significantly affect the flexural behavior of the nanoshells.

5 Conclusions

Based on the two-dimensional nonlocal elasticity theory in conjunction with the first-order shear deformation theory of shells, the dynamic stability of cylindrical nanoshells under combined static and time-dependent periodic axial forces was investigated. The DQM was used to discretize the equations of motion under different boundary conditions in the spatial domain. Afterward, the Bolotin's first approximation method was employed to extract the dynamic instability regions of the cylindrical nanoshells from the obtained system of coupled Mathieu–Hill-type equations in time domain. After validating the approach by exhibiting its fast convergence rate and performing comparison studies with

Fig. 7 a–d The effects of the length-to-thickness ratio on the instability regions of the nanoshells subjected to two different boundary conditions [$h/R = 0.05$, $m = 0$, $\alpha = 0.25$, $\mu = 0.05$ (nm)]: (solid line with square) C–C, (solid line with circle) S–S



existing results, some parametric studies were carried out to present the effects of the small-scale effect parameter, length and thickness-to-mean radius ratios together with different boundary conditions on the principal dynamic instability regions of the cylindrical nanoshells. The results showed that

- By increasing the small-scale parameter, the onset of instable region decreases,
- The greater circumferential wave number m has lower onset of instability region which shows that the dynamic instability occurs at asymmetric mode (i.e., a mode with $m \neq 0$) firstly when the excitation frequency was increased from zero,
- When the dynamic load factor (β) is increased from zero at a given high excitation frequency, the dynamic instability may happen at symmetric mode $m=0$,
- By increasing the length- and thickness-to-mean radius ratio, the bifurcation excitation frequency decreases and increases, respectively.

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